COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-006: Enterprise Load Balancing

Document ID	TS-INF-006
Title	Enterprise Load Balancing
Executive Sponsor	Network Operations, Sr. Manager of Network Services
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Tier	Standard
Version	3.0
Effective Date	07/01/2025
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1. Purpose

OIT is statutorily obligated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy stakeholder needs surrounding Enterprise Load Balancing, this document is submitted as the standard for technology and specifications to provide the service.

The ultimate purpose of this standard is to describe the minimum requirements for designing, installing, securing, monitoring, maintaining, protecting, and decommissioning a load balancing service. Load balancing seeks to: distribute client requests; ensure high availability; meet demand by increasing or decreasing scale; and facilitate continuous integration and delivery.

2. Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Network Operations - Sr. Manager of Network Services

- Owner of Standard; engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Features

- SSL Session Handling: Must provide the ability to terminate SSL sessions or support SSL pass-through; dedicated SSL offloading on hardware-based load balancers.
- Clustering: Must be deployed in a highly available cluster of at least two units, able to sync configurations and session data between members.
- Session Awareness: Must maintain session state with backend services using Layer 3/4 information (sticky sessions) and cookies (session- or time-based).
- Balancing Methods: Must support at minimum round robin and least connections.
- Health Checks: Must provide health checks to verify whether a member server is functioning, with configurable response (e.g., a dedicated health URL).
- Resource Management: Adding and removing member servers without interrupting overall service must be supported, both dynamically and manually.

Design

- The load balancer should have an interface placed in the same VLAN as member servers.
- Services presented by the load balancer must reside behind a firewall.
- The load balancer must not provide indirect access to services that would not otherwise be allowed.
- If used in a production environment, it must also be used in an immediate pre-production test environment (Test/UAT/Staging).

Managing

- There must be a dedicated out-of-band management interface for each member of the load balancer cluster.
- Configurations must be manageable using a GUI and an API (or CLI in lieu of an API).

Monitoring — Required Metrics

- Status of member servers and member server groups.
- Status of overall services presented by the load balancer.
- Number of connections to overall services and to each member server.
- Throughput or data counts to overall services and to each member server.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the "Owner of Standard" listed under Roles and Responsibilities.

Requests may be acquired by completing the "EXCEPTION REQUEST to Technical Standard" form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

- Load balancing (computing) — Wikipedia
- Cloud load balancing — Wikipedia
- NGINX: What Is Load Balancing?
- OSI model — Wikipedia

7. Definitions

API (Application Programming Interface): A set of routines, protocols, and tools for managing a device, specifying how software components should interact.

CLI (Command Line Interface): A means of interacting with a network or computing device via successive lines of text commands.

Cluster: A set of tightly connected load balancers that work together so they can be viewed as a single system.

Cookie: A small piece of data created by a website or load balancer and stored on the user's computer by the browser.

GUI (Graphical User Interface): A human-computer interface using windows, icons, and menus, manipulated by a mouse, trackpad, touch screen, and/or keyboard.

Member server: A single resource behind a load balancer to which connections are given; typically multiple member servers exist for redundancy and scale.

SSL (Secure Sockets Layer): The standard security technology for establishing an encrypted link between a web server and a browser.